



AI in Assent Management

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— BUREAU OF —
RESEARCH AND INNOVATION



We have Data - Need Insights

Today's Reality:

-  Infrastructure inspection reports are huge
-  Hundreds of images, tables, paragraphs
-  Hours to find specific information
-  Manual cross-checking across reports
-  Inspectors in hazardous conditions

Critical Questions That Take Too Long:

- "What defects exist on this component?"
 - "How has corrosion progressed over 5 years?"
 - "When should we intervene?"
 - "Where should we prioritize our budget?"
- We need intelligence, not just information.





The Digital Twin Solution

Digital Twin = 3D Model + Real-Time Sensors + AI Analytics

From Static Reports to Living Intelligence

Traditional	Digital Twin Future
 Static documents	 Continuous updates
 Manual analysis	 AI-powered insights
 Disconnected data	 Integrated intelligence
 Reactive decisions	 Predictive optimization



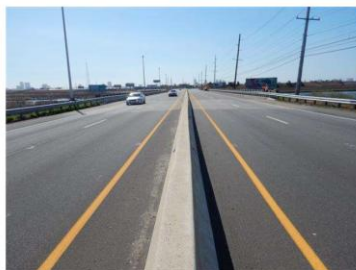
AI-Enabled Bridge Inspection Dashboard



Core Challenges – Bridge Inspection Reports



STATE OF NEW JERSEY
NEW JERSEY DEPARTMENT OF TRANSPORTATION
1035 PARKWAY AVENUE, P.O. Box 600
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BRIDGE RE-EVALUATION SURVEY REPORT

STATE STRUCTURE NO. 0103153
COUNTY STRUCTURE NO.
US 30 OVER DUCK THOROFARE
Atlantic City
ATLANTIC COUNTY

CYCLE No. 23
APRIL 04, 2023

NOTE: This Bridge Re-Evaluation Report
shall be filed immediately after the
22nd Cycle Inspection Report.

Prepared By:



Structural Evaluation and Bridge Management

Single report – 300+ pages

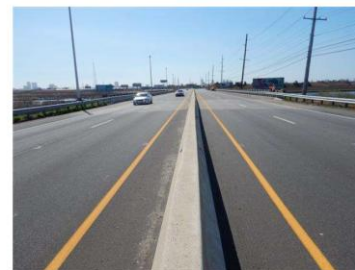
- Long paragraphs
- Numerical records
- Detailed text
- Hundreds of images, tables, paragraphs
- Hours to find specific information

Find something?
Verify details?
Cross-check?
Summary key points?

...



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BRIDGE RE-EVALUATION SURVEY REPORT

STATE STRUCTURE NO. 1102150
COUNTY STRUCTURE NO.
US 1B OVER SHABAKUNK CREEK
Lawrence Township
MERCER COUNTY

CYCLE No. 22
APRIL 19, 2024

NOTE: This Bridge Re-Evaluation Report
shall be filed immediately after the
21st Cycle Inspection Report.

Prepared By:



Structural Evaluation and Bridge Management

How about more...



Customized AI Dashboard for Report Analysis

Why Do We Need a Customized AI Dashboard?



1



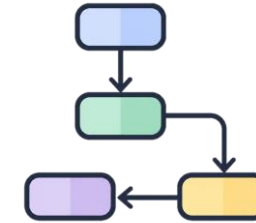
Gemini



Task and Data Adaptation

- Retrieve historical bridge data
- Support better prediction & deeper analysis

2



Trust and Robustness

- Define workflows for infrastructure tasks
- Improve robustness and trustworthiness

3



Cost-Effective Deployment

- One inference costs **>\$1** per 100-page report
- Local deployment reduces inference costs

4

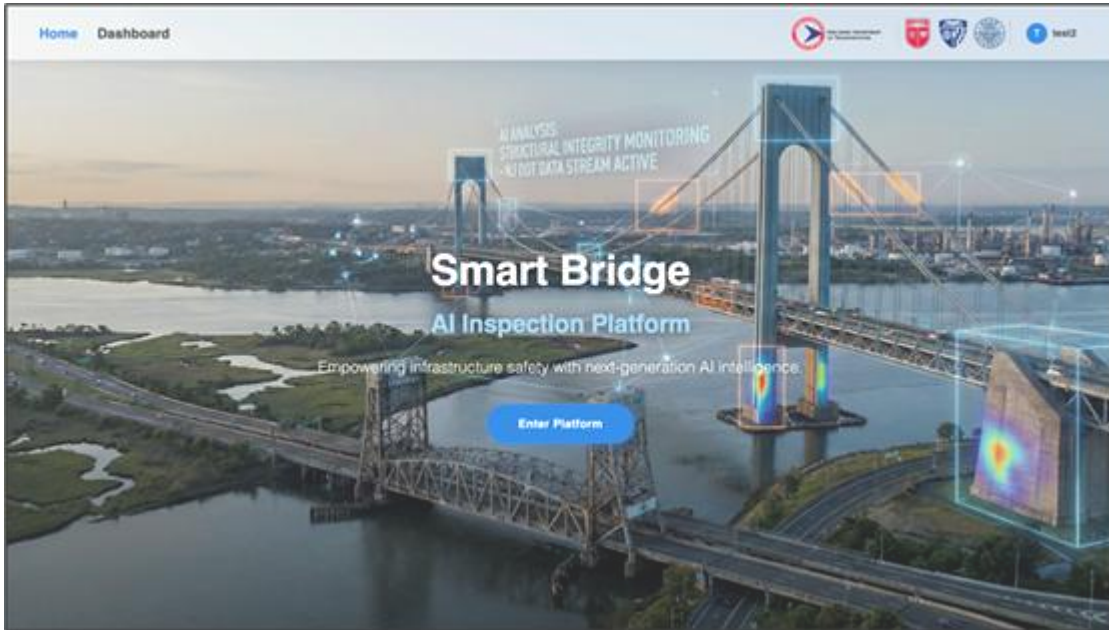


Data Security

- Keep all data on local servers
- Prevent leakage of sensitive information



Smart Bridge Dashboard



1. System Overview

- Core functionalities
- Overall system architecture

2. Dashboard Deployment

- Deployment principles
- Implementation details

3. Performance Evaluation

- Fundamental capability evaluation
- Question response examples



System Overview: Core Functionalities

We designed three types of functions:



1. General Question Answering:

A chatbot that helps users quickly understand documents and compare different documents.

2. Detailed Information Retrieval:

An information retrieval function that helps users quickly locate relevant information within files.

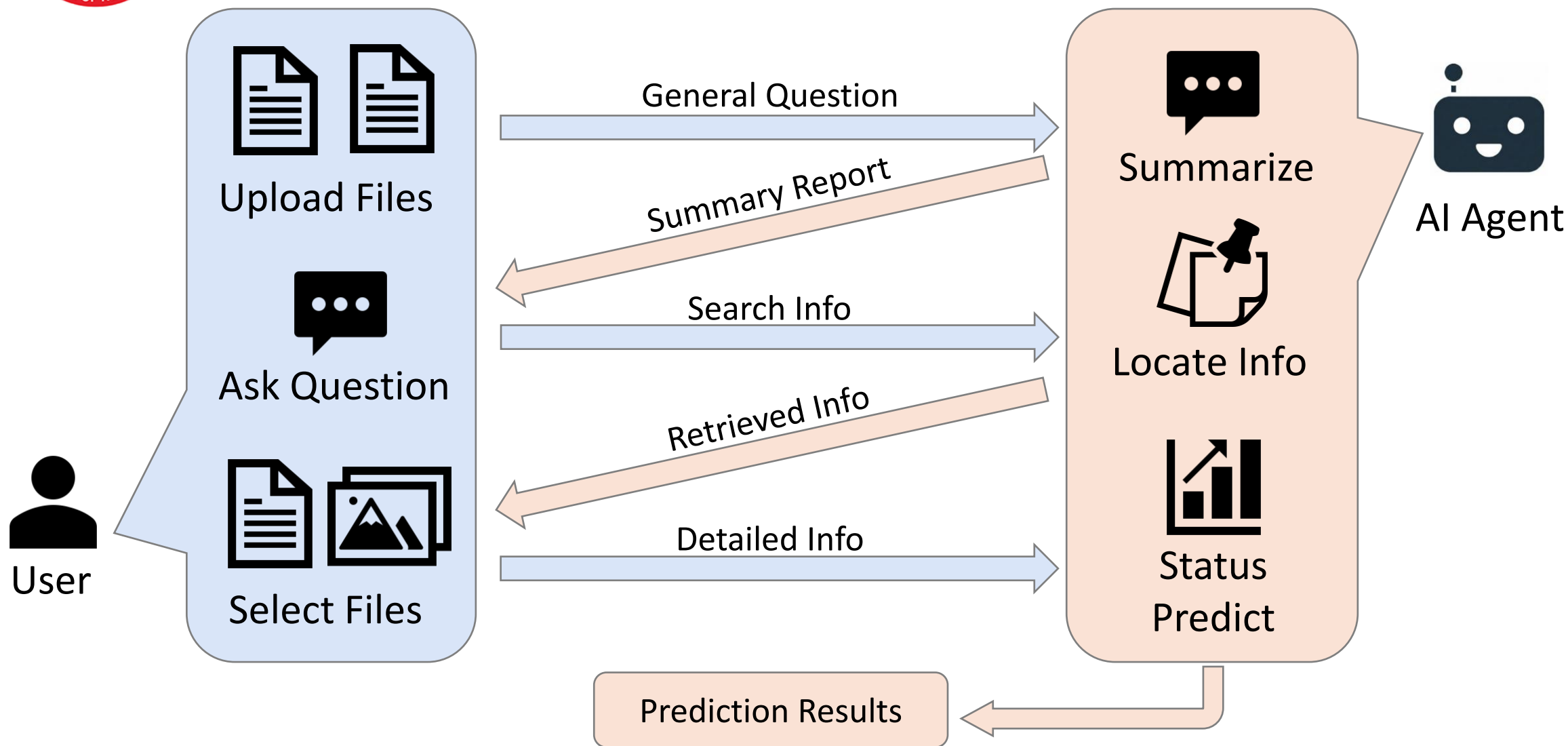


3. Cross-report Information Prediction:

A function that predicts the lifecycle of a component based on image, text, and numerical information.



System Overview: Core Functionalities

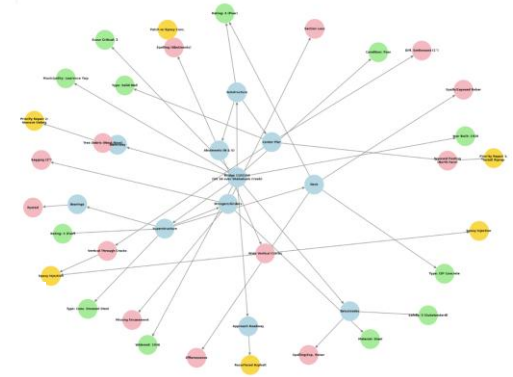




Key Technologies Implemented

1. Knowledge Graph - Enhanced Document Understanding

- Uses a knowledge graph to connect key report information such as damages, locations, repair history.
- Improves document understanding and retrieval accuracy for complex questions.



2. Mixture-of-Experts (MoE) Across LLMs

- Different models demonstrate complementary strengths:
 - Claude: stronger at generating new idea.
 - Gemini: concise and clear responses.
 - GPT: well-structured outputs. ...
- Through our MoE framework, different models leverage their strengths to produce higher-quality and more accurate outputs.





Dashboard Interface

The screenshot displays a web application interface for the NJ DOT AI Assistant. The browser window shows the URL "localhost". The interface has a top navigation bar with "Home" and "Dashboard" tabs, and a user profile "test11". The main content area is divided into a left sidebar and a central chat area. The sidebar includes a "FILES" tab, a "Project Documents" section with a "Click or Drag & Drop" area (supported: PDF, JPG, PNG), and a list of reports: "Report #1102-150" (27.4 MB) and "Report #0103-153" (42.6 MB). The central chat area has tabs for "CHAT ASSISTANT", "DOC RETRIEVAL", and "DEFECT ANALYSIS". The chat interface shows a welcome message from the NJ DOT AI Assistant, a user question "Who are you and how can you help me?", and a detailed response about the assistant's capabilities. A "New Chat" button is in the top right of the chat area, and a "Show Suggestions" button is at the bottom right. A footer note states: "* AI will only analyze selected documents."

Home Dashboard

FILES HISTORY

Project Documents

Click or Drag & Drop
Supported: PDF, JPG, PNG

Report #1102-150
27.4 MB

Report #0103-153
42.6 MB

CHAT ASSISTANT DOC RETRIEVAL DEFECT ANALYSIS

Hello! I am your NJ DOT AI Assistant. I can analyze the files you upload (images, PDFs) and answer your questions.

Who are you and how can you help me?

Simple query detected. Using lightweight model

I am your **NJ DOT Civil Engineering AI Assistant**, specifically designed to provide high-precision, data-driven support for the analysis of bridge inspection reports. I act as a senior engineering consultant to help you manage and interpret structural data in accordance with NJ DOT and FHWA standards.

I can assist you in the following key areas:

1. Inspection Report Analysis

I can extract and summarize critical data from uploaded inspection documents, including:

- **NBI Condition Ratings** (Deck, Superstructure, Substructure, Culvert).
- **Structural Element Assessments** (Quantity and condition state distributions).
- **Inspector Recommendations** and repair priorities.
- **Specific Defect Details** such as spalling, section loss, scour, or delamination.

Ask about the inspection report...

* AI will only analyze selected documents.



ROI

Real Questions, Real Answers:

- "What defects were found on Bridge 405 in last 3 inspections?"
- "Compare steel deterioration rates between bridges in coastal zones"
- "Predict maintenance needs for next fiscal year"

Measurable Impact:

- 🕒 Report analysis: **Hours** → **Minutes**
- 🎯 Search accuracy: **Significantly improved**
- 💰 Cost per analysis: **>\$1** → **pennies** (local deployment)
- ✅ Inspector satisfaction: **High adoption**





AI-Enabled Structure Defect Detection Tool



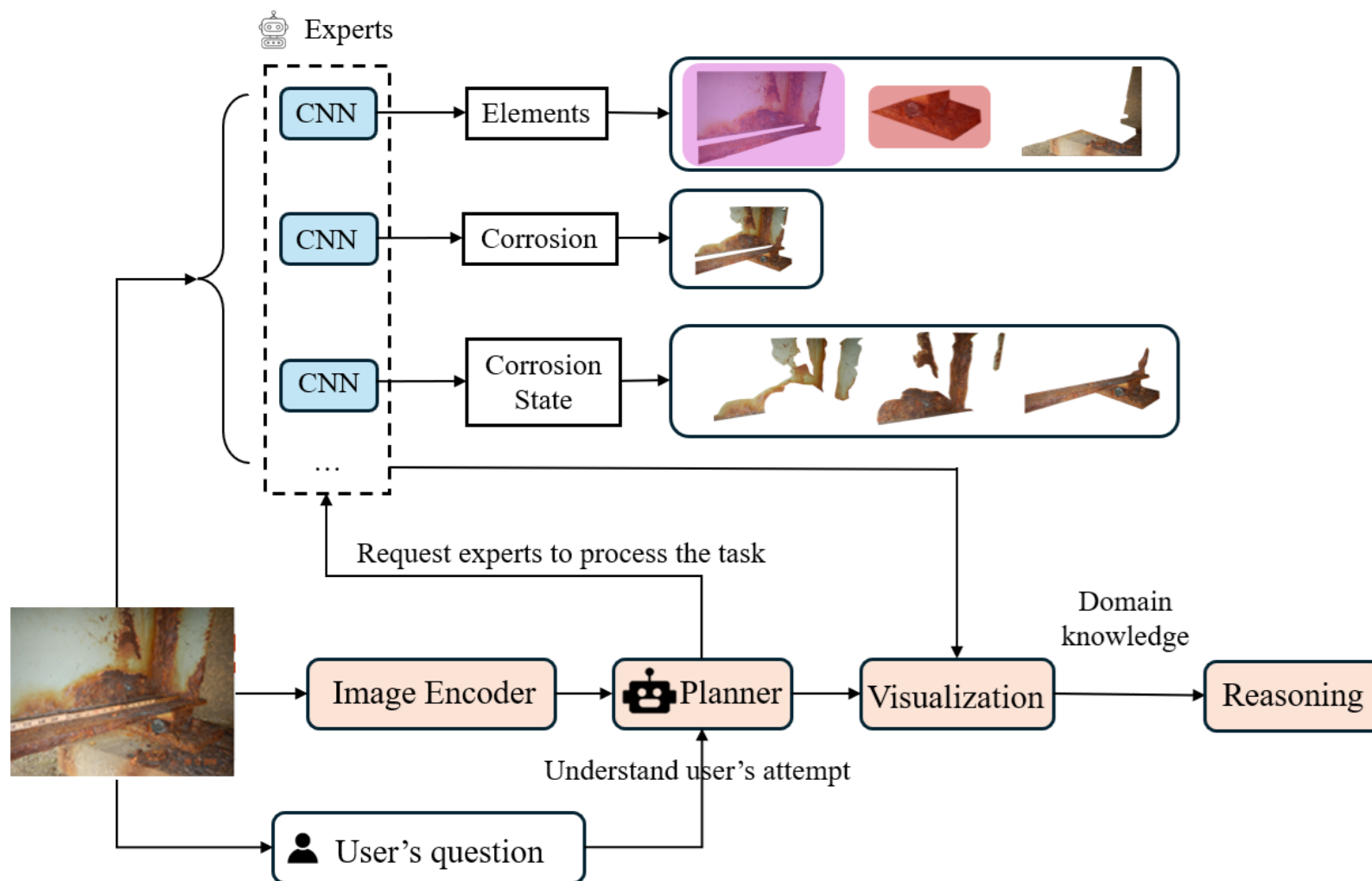
Automated Detection of Bridge Defects

- Bridge deterioration requires timely maintenance and rehabilitation to ensure structural integrity and public safety.
- Traditionally, **manual visual inspection** is labor-intensive, time-consuming, costly, and exposing inspectors to hazardous working environments.
 - ❖ Bridge inspectors take photos and annotate it with written descriptions for each identified defect on structural element.
- We propose an **automated inspection tool** for detecting and quantifying **corrosion defects** from images using natural language inputs.
 - ❖ Photos can be taken by human inspectors or drones
 - ❖ Users upload the image and input queries
 - ❖ Can be extended to multiple defects in different structure types





What is the Tool





Prototype Tool Features

- The tool is based on a novel **hybrid framework** that integrates **vision language models (VLMs)** with domain-specific **convolutional neural networks (CNNs)** to automate detection and evaluation of bridge element defects.
 - ❖ The role definition assigns the VLM a domain-specific professional.
 - ❖ Domain knowledge explicitly grounds the VLM's reasoning within the capabilities of CNN modules, such as structural elements, defects, and corrosion states.
 - ❖ VLM is provided with curated query-response examples covering different levels of query complexity, guiding it to map user questions to the appropriate logical operations.



Image Samples from the Dataset





Sample Questions

Level 1

- Where is the bearing?
- Segment the floor beam.
- Show me the pier.

Level 2

- Where is the bearing and girder?
- Segment the floor beam and bearing.
- Show me the pier and girder.

Level 3

- Show me the rust on bearing.
- Segment the rust on girder of the bridge.
- Segment the rust on pier.

- What is the combined pixel area of rust on the bearing and pier?
- Calculate the total size of corrosion found on the girder and pier in pixels.

Level 4

- Calculate the area distribution of each corrosion condition state.
- What percentage of the image shows Fair, Poor, and Severe corrosion?

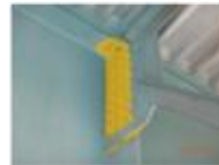
Level 5

- Show me CS2 corrosion on the bearing.
- Highlight CS3 corrosion areas on the pier.

Level 6



Evaluation of Results

Level	L1	L2	L3	L4	L5	L6
Question	Can you highlight the pier for me?	Identify and mask the girder and pier in this bridge image.	Where is the rust on the bearing?	Calculate the total size of corrosion found on the girder and bearing in pixels.	Show me the corrosion severity map.	What parts of the girder have CS3_Poor level corrosion?
Original image						
Ground Truth						
BridgeGPT						
Pure VLM						



Web-based Interface

R BridgeGPT

Upload Bridge Photo

Upload 200MB per file • JPG, PNG, JPEG

Popular Questions

Can you simply describe the figure?

Can you segment all elements?

Show me the deck and the girders

Can you show the rust area for me?

Calculate the corrosion area on pier.

Evaluate the Condition State of the rust on girder.

Ask about bridge logic...

Manage app

Original image



Can you simply describe the figure?

Can you segment all elements?

Can you show the rust on bearing only?

What is the corrosion state of the bearing?

R BridgeGPT

The image shows a close-up view of a bridge bearing, likely made of steel and concrete. There is significant rust and corrosion visible on the steel components of the bearing, as well as on the surrounding structural elements. The concrete substructure appears to be in fair condition with some minor staining.

The image displays a bridge bearing with significant rust and deterioration. AI detection shows the bearing constituting **28.02%** of the total image area, with the girder making up **30.32%** and the pier comprising **19.69%**.



bearing girder pier

Rust is prominently visible on the bearing assembly. The bearing appears to be in poor condition due to the extent of rust. Rust on bearing: **56.12%** of component area.



Rust

The bearing exhibits the following corrosion condition states: **CS2 (Fair): 9.40%** | **CS3 (Poor): 24.40%** | **CS4 (Severe): 12.38%**. Based on the dominant condition state, the overall assessment is severe due to extensive corrosion. As per AASHTO Bridge Element Inspection Manual, Section 3.1.13 Bearings, particularly for Elastomeric Bearing Element #310, the severe condition state for corrosion indicates alignment changes that may impede load carrying capacity. Therefore, immediate rehabilitation of the bearing is recommended.



CS2_Fair CS3_Poor CS4_Severe



Thank you.

Questions?